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Fractions and Decimals in Equations

Real situations often involve more difficult numbers. The examples below demonstrate useful tips for working with fractions and decimals.

Although you can solve problems using fractions, most problems are easier to solve without them. To eliminate the fractions from a problem, multiply both sides of the equation by ~~the reciprocal of~~ a common denominator.

Example 1

Hide

$$\frac{1}{4} + \frac{5}{2}x = \frac{9}{4}x - 2$$

Multiply both sides by 4, or $\frac{4}{1}$.

$$\frac{4}{1} \times \left(\frac{1}{4} + \frac{5}{2}x \right) = \frac{4}{1} \times \left(\frac{9}{4}x - 2 \right)$$

Use the distributive property.

$$\left(\frac{4}{1} \times \frac{1}{4} \right) + \left(\frac{4}{1} \times \frac{5}{2}x \right) = \left(\frac{4}{1} \times \frac{9}{4}x \right) - 4(2)$$

Finish by using inverse operations.

$$\begin{aligned} 1 + 10x &= 9x - 8 \\ 1 + x &= -8 \\ x &= -9 \end{aligned}$$

The solution is $x = -9$

$$\begin{aligned} 1 + 10x &= 9x - 8 \\ 1 + 10x - 9x - 1 &= 9x - 8 - 9x - 1 \\ x &= -9 \end{aligned}$$

You can also eliminate the decimals from an equation. Find the decimal with the greatest number of decimal places. Multiply each term by a power of ten with that same number of zeros.

* Remember, when multiplying by a power of ten, the number of zeros in the power of ten dictates how many places to move the decimal point to the right. *

Example 2

Hide

$$5.22 + 0.6n = 1.5n$$

The term 5.22 has two decimal places. Multiply both sides by 100 to eliminate the decimals.

$$5.22 + 0.6n = 1.5n$$

$$100 \times (5.22 + 0.6n) = 100 \times 1.5n$$

Use the distributive property.

$$(100 \times 5.22) + (100 \times 0.6n) = 100 \times 1.5n$$

Finish by using inverse operations.

$$\begin{aligned} 522 + 60n &= 150n \\ 522 &= 90n \\ 5.8 &= n \end{aligned}$$

The solution is $n = 5.8$

$$\begin{aligned} 522 + 60n &= 150n \\ 522 + 60n - 60n &= 150n - 60n \\ 522 &= 90n \\ \frac{522}{90} &= \frac{90n}{90} \\ 5.8 &= n \end{aligned}$$

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Lesson 3: Solving Equations

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